



Course Name: Artificial Intelligence & Neural Networks	
Semester V	
Course Code:	BCA-DC501
Course Title:	Artificial Intelligence & Neural Networks
Number of Credits:	Total: (L: 3 T:1)
Total Number of teaching hours	40 Hours
Pre-requisites	
Course Category:	Core

Rationale:

Artificial Intelligence is utilized heavily in computizing cognitive functions such as speech and Vision. Often these functions are achieved through the use of Neural networks. In this module, we will study very popular Neural Network architectures for achieving various cognitive functions such as Object recognition, natural language processing besides explores reinforcement learning. We will study and practice various use cases ranging from text generation, object tagging to fraud detection and learning games.

Learning Outcome:

- Students will learn the basic concepts and techniques of Artificial Intelligence. They should be able to develop AI algorithms for solving practical problems.

Detailed Contents

Unit	Title	Hours
Unit 1	Foundations for AI	10
1	Defining AI techniques, Comparison - AI, ML and Deep Learning, Artificial Intelligence and its applications, Artificial Intelligence Techniques, Level of models, criteria of success, Intelligent Agents, Nature of Agents, Learning Agents. AI Techniques, advantages, and limitations of AI, Impact and Examples of AI, Application domains of AI.	
Unit 2	Problem solving techniques & Logic	10
2	State space search, control strategies, heuristic search, problem characteristics, production system characteristics., Generate and test, Hill climbing, best first search, A* search, Constraint satisfaction problem, Mean-end analysis, Min-Max Search, Alpha-Beta Pruning. Propositional logic, predicate logic, Resolution, Resolution in propositional logic and predicate logic, Clause form, unification algorithm	
Unit 3	Convolution Neural Networks	10

3	Image classification, Text classification, Image classification and hyper-parameter tuning, Emerging NN architectures: ResNet, AlexNet – Applications	
Unit 4	Recurrent Neural Networks	10
4	Recurrent Neural Networks, Building recurrent NN, Long Short-Term Memory, Time Series Forecasting, Bidirectional RNNs, Encoder-decoder sequence to sequence architectures - BPTT for training RNN, Computer Vision - Speech Recognition - Natural language Processing, Case studies in classification, Regression and deep networks.	10

Learning Approach:

- To understand Principles of Data Warehousing and Data Mining.
- To understand Architecture of Data Warehouse.
- To understand Various Data preprocessing Methods.
- To understand Data Mining techniques.

References/suggested learning resources:

TextBook and Learning Resource:

- A Classical Approach to Artificial Intelligence, M.C. Trivedi, Khanna Book Publishing, 2019.
- Artificial Intelligence: A modern approach by Stuart Russel, Pearson Education, 2010.
- Artificial Intelligence by Rich and Knight, The McGraw Hill, 2017.
- Artificial Intelligence: A new synthesis by Nils and Nilson, Elsevier, 1997.
- Artificial Intelligence by Luger, Pearson Education, 2002.
- Artificial Intelligence by Padhy, Oxford Press, 2005.
- Ian Goodfellow, Yoshua Bengio, and Aaron Courville, “Deep Learning”, First Edition, MIT Press, 2016.
- Nikhil Buduma and Nicholas Lacascio, “Fundamentals of Deep Learning”, First Edition, O.Reilly, 2017
- Russel,S., and Norvig,P., (2015), Artificial Intelligence: A Modern Approach, 3rd Edition, Prentice Hall

Reference Books:

- E. RICH, K. KNIGHT, S. B. NAIR (2017), Artificial Intelligence, McGraw Hill Education, 3rd Edition.
- R.S. SUTTON, A.G. BARTO (2015), Reinforcement Learning: An Introduction, The MIT Press, 2nd Edition.
- Josh Patterson, Adam Gibson "Deep Learning: A Practitioner's Approach", O'Reilly Media, 2017
- Laura Graesser, Wah Loon Keng "Foundations of Deep Reinforcement Learning: Theory and Practice in Python" Addison-Wesley Professional -2020
- Jon Krohn, Grant Beyleveld, Aglaé Bassens "Deep Learning Illustrated: A Visual, Interactive Guide to Artificial Intelligence", 1st edition Addison-Wesley Professional 2019

(b) Open source software and website address:

- <https://www.edx.org/course/artificial-intelligence-ai>
- <https://www.udemy.com/course/artificial-intelligence-az/>
- Department of Computer Science, University of California, Berkeley, <http://www.youtube.com/playlist?list=PLD52D2B739E4D1C5F>

- An Introduction to Artificial Intelligence By Prof. Mausam, IIT Delhi
- Artificial Intelligence, Prof. Sudeshna Sarkar, IIT Kharagpur

Tutorial List:

- Numerical type questions on CNN-
 - a. Parameters tuning
 - b. Convolution function
 - c. Different types of filters
- Fuzzy Logic and Neural Networks

Course Name:	
Semester V	
Course Code:	BCA-DC502
Course Title:	Machine Learning Using Python
Number of Credits:	Total: (L: 2 T:1 P: 2)
Total Number of teaching hours	40 Hours
Pre-requisites	Programming in Python
Course Category:	Core

Rationale:

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcome:

At the end of the course, the student will be able to:

- i. Use python libraries for performing machine learning on datasets.
- ii. Understand and apply the machine learning workflow on datasets.
- iii. Implement data preprocessing on datasets.
- iv. Implement machine learning algorithms.
- v. Determine the performance of machine learning algorithms and evaluate the models.

Detailed Contents

Unit	Title	Hours
Unit I	Essential Python libraries for Machine Learning and Introduction to Machine Learning	8
	Essential Python libraries for Machine Learning: Numpy: Creation of Numpy arrays, array indexing, slicing, reshaping, array math, assignment, Manipulating Tabular Data using Pandas: Pandas Series and DataFrame, Data Visualization using matplotlib, seaborn: Basics, line charts, bar charts, pie charts, scatter plots, subplots, Scikit-learn: Introduction, Getting the datasets, Features and Applications Introduction to Machine Learning: Basics, Significance, Problems ML can solve, Classification of Machine Learning Techniques	
Unit 2	Machine Learning Workflow, Data Preprocessing and Supervised Learning Techniques	14
	Machine Learning Workflow and Data Preprocessing: Machine Learning Lifecycle, Types of Data in datasets, Dataset Loading, Data Preprocessing: Data Outlier analysis, treating for missing values, encoding categorical data, splitting the dataset into training and test dataset, feature scaling, Feature Selection Techniques: Filter, Wrapper and Embedded methods. Supervised Learning Techniques: Introduction, Classification of supervised learning techniques, Linear Regression, Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machines, Naive Bayes Algorithm, Decision Tree, Model evaluation - accuracy, recall, precision, F1 score and Confusion matrices. Implementation of the learning techniques and determination of performance.	
Unit 3	Unsupervised Machine Learning Techniques	8
	Unsupervised Learning Techniques: Approaches: Clustering, Association, Dimensionality reduction, K-means, Hierarchical clustering, Principal Component Analysis, Frequent Itemset Mining, Apriori algorithm, Implementation of the learning techniques and determination of performance.	
Unit 4	Ensemble Methods, Reinforcement Learning & Case studies on Machine Learning	8

	Ensemble Methods & Reinforcement Learning: Bagging, Boosting, Random Forest, Reinforcement Learning: RL Framework and TD Learning Case Studies on Machine Learning: Analyzation on datasets using ML workflow.	
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List Of Practical:

Practical based on:

- i. Python libraries for machine learning
- ii. Dataset loading
- iii. Data preprocessing
- iv. Machine learning algorithms
- v. Dataset analysis using ML techniques.
- vi. Performance prediction

Learning Approach:

To learn & apply the fundamentals of python libraries for machine learning, machine learning workflow, data preprocessing, data analysis using ML algorithms.

References/suggested learning resources:

1. Python Machine Learning, Wei-Meng Lee, Wiley (2019)
2. Introduction to Machine Learning with Python, Andreas C. Müller, Sarah Guidoy, O'Reilly Media, Inc. (2016)
3. Machine Learning for Absolute Beginners, Oliver Theobald, Scatterplot Press (2021)
4. Machine Learning with Python Cookbook, Chris Albon, O'Reilly Media, Inc. (2018)
5. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, Third Edition, Aurélien Géron, Shroff/O'Reilly (2022)
6. Fundamentals of Machine Learning for Predictive Data Analytics: Algorithms, Worked Examples, and Case Studies by John D. Kelleher, Brian Mac Namee, and Aoife D'Arcy, MIT Press (2015)

Course Name: Mobile Application Development	
Semester V	
Course Code:	BCA-DC503
Course Title:	Mobile Applications Development
Number of Credits:	Total: (L: 2 T:1 P: 2)
Total Number of teaching hours	40 Hours
Pre-requisites	NIL
Course Category:	Core

Rationale:

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcome:

- Student knows mobile devices and mobile platforms
- Student has knowledge concerning mobile operating systems and their architecture
- Student is familiar with wireless communications standards and data transmission standards
- Student knows how to prepare a mobile application for distribution In terms of skills.
- Student can recognize and setup a mobile device and application runtime environment
- Student is able to setup programming tools for a mobile application developer (for selected modern mobile platforms)
- Student can select appropriate data transmission standards In terms of social competence.
- Student understands the need for continuous improvement of his/her skills due to the rapidly changing environment of mobile devices.

Detailed Contents

Unit	Title	Hours
Unit 1	Introduction to Web Application Development	10
1.1	Overview of Internet and web, HTML Tags, Forms & Frames , Introduction to Java Script and Cascading Style Sheets, DHTML.	

1.2	Introduction to PHP, MySql/MariaDB, Introduction to Web Server, Installation & Configuration of WAMP/LAMP/XAMPP, IIS. Developing Web Applications in PHP with Databases, Introduction to BootStrap.	
Unit 2	Android Systems	10
2.1	Android Systems: Introduction to Mobile devices and applications, Open Handset Alliance (OHA), Overview of Android OS and architecture, installing android studio, understanding the android build system.	
2.2	Introduction to Android Application Components: Activity, service, content provider and broadcast received, Intents, Receiving and Broadcasting Intents, Android Manifest File and its common settings, Using Intent Filter, Permissions. Activities and Intents: Declaring and starting a new activity, switching between activities, passing data and returning value from activity, state of activity, storing persistent activity data, understanding activity life cycle, Types of intents, intent filter, context, data sharing using intent.	
Unit 3	Android Web Application Development	10
3.1	Android User Interface: Basic android UI, android layouts, view attributes, Objects & Controls and uses UI events and event listeners, animations, notifications, progress dialog, Action bar, toolbar, menus and pop ups, Floating action button, Tab based UI,	
3.2	Fragment, Types of Fragment, Fragment Lifecycle, communication between fragment and activity, add, replaced remove fragment, Listview, material design, recyclerview.	
Unit 4	Android Storage and API	10
4.1	Android Storage: Using Android Data and Storage APIs, Managing data using SQLite, Sharing Data between Applications with Content Providers Android APIs: Multimedia, Using Android Networking APIs, Using Android Web APIs, Using Android Telephony APIs, android location based services	

List of Practical:

1. Installation of Android studio.
2. Development Of Hello World Application
3. Create an application that takes the name from a text box and shows hello message along with the name entered in text box, when the user clicks the OK button
4. Create a screen that has input boxes for User Name, Password, Address, Gender(radio buttons for male and female), Age (numeric), Date of Birth (Date Picket), State (Spinner) and a Submit button. On clicking the submit button, print all the data below the Submit Button (use any layout)
5. Design an android application to create page using Intent and one Button and pass the Values from one Activity to second Activity

6. Design an android application Send SMS using Intent
7. Create an android application using Fragments
8. Design an android application Using Radio buttons
9. Design an android application for the menu.
10. Create a user registration application that stores the user details in a database table.

Learning Approach:

1. Design scripts to meet given interface requirements.
2. Use variables, properties and other code elements appropriately to implement the code design.
3. Devise, carry out and evaluate functional test strategies of mobile design.
4. Implement and evaluate techniques for the installation of mobile applications and delivery via various channels.

References/suggested learning **resources:**

(a) Books

- 1. Abraham Silberschatz and Peter Baer Galvin, "Operating System Concepts", Addison-Wesley
- 2. Milan Milankovic, "Operating Systems, Concepts and Design", Tata McGraw-Hill.
- 3. Harvey M Deital, "Operating Systems", Addison Wesley
- 3. Richard Peterson, "Linux: The Complete Reference", Osborne Tata McGraw-Hill.
- 4. Sibsankar Halder and Alex A Aravind, "Operating Systems", Pearson Education
- 5. D M Dhamdhare, "Operating Systems : A Concept based Approach", McGraw Hill.
- 6. Charles Crowley, "Operating Systems: A Design-Oriented Approach", Tata McGraw Hill Education".
- 7. Stuart E. Madnick & John J. Donovan. Operating Systems. McGraw Hill.

Course Name: Software Project Management & SPM Tools	
Semester V	
Course Code:	BCA-DC504
Course Title:	Software Project Management & SPM Tools
Number of Credits:	Total: (L: 2 T:1 P: 2)
Total Number of teaching hours	32 Hours
Pre-requisites	Basic knowledge of Software Engineering
Course Category:	Core

Rationale:

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcome:

- Understand the concept of Software Project Management
- To provide fundamental knowledge of software project management and corresponding software tools.

Detailed Contents

Unit	Title	Hours
Unit 1	Software Project Management Concepts	08
1	Introduction, Project and Software project, Software project vs other project, Importance and Problems in software project management, Process of SPM. Characteristics of good project manager, Successful Software Project Manager, Overview of Software Project Planning.	
Unit 2	Software Project Scheduling & Cost Estimation	10
2	Objectives of activity planning, Work breakdown structure, Network planning model: Critical path method (CPM), Program evaluation and review technique (PERT), Precedence diagramming method (PDM), Shortening project duration, Identifying critical activities. Software Effort Estimation: Basis of software Estimating, Software effort estimation techniques, expert Judgment, Estimating by analogy. Bottoms-up estimating, Top-down approach and parametric models.	
Unit 3	Project Evaluation & Risk Management	10

3	Project Evaluation: Strategic Assessment, Technical Assessment, cost-benefit analysis, Risk Evaluation. Selection of Appropriate Report, Project approach: Choosing technologies, choice of process models, structured methods. Risk Identification, Planning, Evaluation and Management, Categories of Risk, Framework for dealing with risk, evaluating Risks to the schedule.	
Unit 4	Software Quality & Configuration Management	12
4	TQM, Six Sigma, SEI CMM , McCall Models for Software Quality , Defining and importance of software quality, Concept, Requirement and Elements of SCM, Baseline, SCM Repository, Versioning and version control, SCM Process, Change Control Process. Configuration Audit and Status Reporting. Study of Software Project Management tools .	

List Of Practical:

Laboratory works should be done covering all the topics listed above and a small work should be carried out using the concept learnt in each unit in the group. Work should be assigned on an individual basis. Students may choose project Management tools like (MS Project, OpenProj, dot Project, Trello, Asana, ClickUp etc).

Learning Approach:

This course provides the comprehensive knowledge about Software Project Management, which encompasses with Software Project Planning, Scheduling, Cost Estimation, Risk management, Quality management and Configuration management

References/suggested learning resources:

1. Cotterell, B. H. (2018). Software Project Management. McGraw-Hill
2. Dutt, S. C. (n.d.). Software Project Management. Pearson Education India.
3. A.S. Kelkar (Node). Software Project Management. PHI Learning.

Course Name: Advance Java Programming	
Semester V	
Course Code:	BCA-EC501
Course Title:	Advance Java Programming
Number of Credits:	Total: (L: 2 T:1 P: 2)
Total Number of teaching hours	40 Hours
Pre-requisites	NIL
Course Category:	Elective

Rationale:

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcome:

- Student knows java applications in web development
- Dynamic web pages using Servlets
- Session management
- Designing custom tags using JSP
- Data access, searches and queries using HQL
- Web Application development using Spring framework
- Handling crosscutting concerns using Spring AOP

Detailed Contents

Unit	Title	Hours
Unit 1	Introduction to J2EE	10

1.1	Distributed Web Application Multi-Tier Architecture, MVC architecture, HTTP Basics, Introduction to servlets and its life cycle, problems with CGI/Perl interface, generic and http servlet, servlet configuration,	
1.2	Various session tracking techniques, servlet context, servlet configuration, servlet listeners, filters and annotations	
Unit 2	JSP Basics and Architecture & Struts Framework:	10
2.1	JSP directives, Scripting elements, standard actions, implicit objects, JSP design strategies. Using JSP Beans, Expression languages, JSP custom tags	
2.2	Introduction of Struts and its architecture, advantages, and application of Struts.	
Unit 3	Hibernate Framework:	10
3.1	Comparison between JDBC and Hibernate, Principles of Object Relational Mapping, Hibernate configuration, mapping and annotations,	
3.2	Session management, searches and queries using HQL, Transaction integration and demarcation.	
Unit 4	Spring Framework:	10
4.1	Introduction of Spring Framework: Configuration of Spring environment, Spring Architecture, Dependency Injection,	
4.2	Inversion of Control (IoC), Spring Boot, Using Hibernate with Spring, Spring AOP and Spring Web.	

List of Practical:

As per the Subject teacher assigned as per theory curriculum.

Learning Approach:

1. Model Java EE architectural components, Servlet creation and session management for web applications.
2. Inspect the fundamentals of Java Server Pages (JSP) and Struts for web-based applications.
3. Appraise Hibernate Framework of JEE and apply constructs of Object Relational Mapping.
4. Elaborate principles of Dependency Injection and its application in JEE.
5. Design applications based on Spring Boot and Spring AOP

1. (a) Books

TEXT BOOKS:

TB1. John Hunt and Chris Loftus, "Guide to J2EE: Enterprise Java", Springer Verlag Publications, 1st Edition, 2003.

TB2. Govind Sesadri, "Enterprise Java Computing: Application and Architectures", Cambridge University Publications, 2nd Edition, 1999.

TB3. Jeff Linwood and Dave Minter, "Beginning Hibernate", Apress Publishing Co., 2nd Edition, 2010.

TB4. Rod Johnson, "Professional Java Development with the Spring Framework", John Wiley & Sons, 2nd Edition, 2005.

REFERENCE BOOKS:

RB1. Ted Neward, "Effective Enterprise Java", Pearson Education, 2nd Edition, 2004.

RB2. Jim Farley and William Crawford, "Java Enterprise in a Nutshell", O'Reilly and Associates, 3rd Edition, 2005.

RB3. James Holmes and Herbert Schildt, "The Complete Reference- Struts", Tata McGraw Hill, 2nd Edition, 2007.

Course Name: Data warehouse and Data mining	
Semester V	
Course Code:	BCA-EC502
Course Title:	Data Warehouse and Data mining
Number of Credits:	Total: (L: 2 T:1 P: 2)
Total Number of teaching hours	32 Hours
Pre-requisites	Basic knowledge of DBMS
Course Category:	Core

Rationale:

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcome:

- Understand the concept of Data Mining

Detailed Contents

Unit	Title	Hours
Unit 1	Data mining and Data preprocessing	10
1	Data mining: Introduction, Data mining – on what kind of data, data mining functionalities – what kind of patterns to be mined, Classification of data mining systems, data mining task primitives, integration of a data mining systems with a database or data warehouse systems, major issues in data mining. Data preprocessing: Descriptive data summarization, data cleaning, data integration and transformation, data reduction, data discretization and concept hierarchy generation	
Unit 2	Data warehouse and OLAP technology	10
2	Data warehouse and OLAP technology: What is data warehouse, A multidimensional data model, data warehouse architecture, data warehouse implementation, data warehouse usage, OLAP, OLAM Mining frequent patterns, association and correlation, efficient and scalable frequent itemset mining methods, From association mining to correlation analysis	
Unit 3	Classification and Cluster Analysis	10

3	Classification and prediction: Introduction, issues, classification by decision tree induction, rule based classification, classification by back propagation, lazy learners, other classification methods, Prediction: accuracy and error measures, evaluating the accuracy of a classifier or predictor. Cluster Analysis: Types of data in cluster analysis, a categorization of major clustering methods, partitioning methods.	
Unit 4	Mining complex types & Applications and trends in data mining	10
4	Mining complex types of data: Multidimensional analysis and descriptive mining of complex data objects, mining spatial database, multimedia database, mining world wide web. Applications and trends in data mining: Data mining applications, data mining system products and research prototypes, social impact of data mining, trends in data mining.	10

List Of Practicals :

- Practical based on demonstrate various OLAP operation on multi-dimensional data
- Practical Based on Apply data mining algorithms on real time data
- Practical Based on Supervised & Un Supervised Learning on Real Time Problems.
- Practical Based on Assess most appropriate data mining algorithm for given situation

Learning Approach:

- To understand Principles of Data Warehousing and Data Mining.
- To understand Architecture of Data Warehouse.
- To understand Various Data Preprocessing Methods.
- To understand Data Mining techniques.

References/suggested learning resources:

Text Book and Learning Resource:

- Data Warehousing (Fundamentals for IT Professionals) : Paulraj Ponniah
- Data Mining and WareHousing : M.SudheepElayidom
- Data Mining : Vikram Pudi, P. Radha Krishna
- Data Warehousing : Reema Thareja

Reference Books :

- Margaret Dunham, “ Data Mining: Introductory and Advanced Topics, 1/e”, Pearson
- G. K. Gupta, “Introduction to Data Mining with Case Studies”, PHI, 2006.
- W. H. Inmon, “Building the Operational Data Store”, 2nd Ed., John Wiley, 1999
- B. M. Shawkat Ali, Saleh A. Wasimi, “Data Mining Methods and Techniques”, Cengage Learning, 2010

(b) Open source software and website address:

Course Name: ERP & ERP Tools	
Semester V	
Course Code:	BCA-EC503
Course Title:	ERP & ERP Tools
Number of Credits:	Total: (L: 2 T:1 P: 2)
Total Number of teaching hours	40 Hours
Pre-requisites	Basic knowledge of Business & Information System
Course Category:	Core

Rationale:

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcome:

At the end of the course, the student will be able to:

- Understand the ERP Fundamentals.
- Understand ERP Functional Modules .
- Understand ERP Solutions..
- Understand ERP Implementation .

Detailed Contents

Unit	Title	Hours
Unit 1	Introduction to ERP	08
	Introduction: Overview of enterprise systems ñ Evolution - Risks and benefits - Fundamental technology - Issues to be considered in planning design and implementation of cross functional integrated ERP systems .	
Unit 2	ERP Solutions and Functional Modules	10
	ERP Solutions and Functional Modules: Overview of ERP software solutions- Small, medium and large enterprise vendor solutions, BPR and best business practices - Business process Management, Functional modules.	
Unit 3	ERP Implementation	10

	ERP Implementation: Planning Evaluation and selection of ERP systems - Implementation life cycle - ERP implementation, Methodology and Framework- Training ñ Data Migration - People Organization in implementation-Consultants, Vendors and Employees.	
Unit 4	Post Implementation & Emerging Trends in ERP:	12
	Post Implementation: Maintenance of ERP- Organizational and Industrial impact; Success and Failure factors of ERP Implementation. Emerging Trends on ERP: Extended ERP systems and ERP add-ons -CRM, SCM, Business analytics - Future trends in ERP systems-web enabled, Wireless technologies, cloud computing.	

List Of Practical:

- Introduction to ERP System and Review on different ERP packages.
- Study on Odoo ERP package.
- Identification and brief introduction of different Modules in Odoo ERP package.
- Detail study of odooerp modules.
- Identify different processes of the odooerp modules.
- Study on odooerp business functions

Learning Approach:

- To know the basics of ERP To understand the key implementation issues of ERP
- To know the business modules of ERP
- To be aware of some popular products in the area of ERP
- To appreciate the current and future trends in ERP

References/suggested learning resources:

1. Alexis Leon, ERP demystified, second Edition Tata McGraw-Hill,2008.
2. Sinha P. Magal and Jeffery Word, Essentials of Business Process and Information System, Wiley India,2012
3. Jagan Nathan Vaman, ERP in Practice, Tata McGraw-Hill,2008
4. Alexis Leon, Enterprise Resource Planning, second edition, Tata McGraw-Hill,2008.
5. Mahadeo Jaiswal and Ganesh Vanapalli, ERP Macmillan India,2009
6. Vinod Kumar Grag and N.K. Venkitakrishnan, ERP- Concepts and Practice, PHI,2006.

Course Name: Programming in R	
Semester V	
Course Code:	BCA-EC504
Course Title:	Programming in R
Number of Credits:	Total: (L: 2 T:1 P: 2)
Total Number of teaching hours	40 Hours
Pre-requisites	
Course Category:	Core

Rationale:

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcome:

At the end of the course, the student will be able to:

- Understand the R Fundamentals.
- Write R scripts.
- Apply R data structure, functions and OOPS concepts.
- Use R to handle the data, manipulate and visualize.
- Use R for performing statistics on data and design GUI.

Detailed Contents

Unit	Title	Hours
Unit 1	R Fundamentals	10
	Fundamentals: Features of R, Significance and advantages over other programming language, Literals, Variables, Operators in R, Data types, Operators: Arithmetic, Assignment, Logical, Relational, Miscellaneous, Keywords Decision Making and Looping: if statement, if...else statement, switch statement, if...else Ladder, ifelse() function, for loop, while Loop, repeat Loop, break Statement, next Statement, nested loops.	
Unit 2	R Data Structures, Functions and Object-Oriented Programming	10
	R Data Structures: Scalars, Vectors, Matrices, List, Dataframes, Matrices, Arrays, Factors, Creation, Access, and Manipulation of the data structures R Functions: Types, user defined function, variable scope.	

	R Objects & Class Inheritance: Object, Class: S3, S4, Reference Classes, Inheritance Types.	
Unit 3	Data Handling and Visualization in R	14
	Importing & Manipulating Data: Reading Tabular Data files, Reading CSV files, Importing data from excel, Analyzing CSV and Excel files, Accessing database, Saving in R data, Loading R data objects, Writing to files, Selecting rows/observations, Selecting columns/fields, Merging data, Converting variable types, Data sorting, Data aggregation Data Visualization: Bar graph, Histogram chart, pie graph, box chart, strip chart, Plot Function, Sub Plot Saving plot, Color in Graphs	
Unit 4	Statistics for Data Science & Designing GUI	6
	Statistics for data science: Mean, Median, Mode, Standard Deviation, Variance, Data Analysis Designing GUI: Building interactive application and connecting it with database.	

List Of Practical:

Practical based on:

- i. R Fundamentals – operators, data types.
- ii. Decision Making and Looping.
- iii. R data structures, functions, OOPS concepts.
- iv. Reading Tabular, CSV and Excel files, accessing database, manipulating and visualizing data.
- v. Statistics and GUI programming using R.

Learning Approach:

To learn & apply the fundamentals of R programming, data structures, OOPS concepts, handling data, statistics, analyzing, visualizing and performing GUI programming.

References/suggested learning resources:

1. Andrie de Vries, Joris Meys, R for Dummies A Wiley Brand, 2nd Edition, John Wiley and Sons, Inc, 2015, ISBN-13 : 978-1119055808
2. Garrett Grolmund, Hands on Programming With R: Write Your Own Functions and Simulations, Shroff/O'Reilly, 2014
3. Sandip Rakshit, R Programming for Beginners, McGraw Hill Education (2017)
4. JD Long et al., R Cookbook: Proven Recipes for Data Analysis, Statistics, and Graphics, O'Reilly Media (2019)
5. Richard Cotton, Learning R: A Step-by-Step Function Guide to Data Analysis, Shroff/O'Reilly; First Edition (2013)
6. Mark Gardener, Beginning R: The Statistical Programming Language, Wiley (2013)
7. Michael Lawrence et. al. Programming Graphical User Interfaces in R, Chapman and Hall/CRC (2012)
8. Dr. Rob Kabacoff, R in Action: Data Analysis and Graphics with R, Manning (2015)

(b) Open source software and website address:

Course Name: IoT Application Development	
Semester V	
Course Code:	BCA-EC505
Course Title:	IoT Application Development
Number of Credits:	Total: (L: 2 T:1 P:2)
Total Number of teaching hours	40 Hours
Pre-requisites	
Course Category:	Core

Rationale:

The purpose of this course is to impart knowledge on IoT Architecture and various protocols, study their implementations and various applications of IoT.

Learning Outcome:

- Exemplify the IoT concepts, Physical Design, Logical Design, Enabling Technologies, IoT Challenges, IoT Levels, Applications, IoT Components, Layers and Protocols, Use cloud for IoT applications. Interface embedded boards & program them Develop IoT applications

Detailed Contents

Unit	Title	Hours
Unit 1	IoT Introduction	8
1	Introduction to IoT: Definition & Characteristics of IoT, History and Evolution of IoT, IoT-An Architectural Overview Building an architecture, Main design principles and needed capabilities, An IoT architecture outline, standards considerations. M2M and IoT Technology Fundamentals- Devices and gateways, Local and wide area networking, Data management, Business processes in IoT, Everything as a Service (XaaS), M2M and IoT Analytics, Knowledge Management	
Unit 2	IoT Data Communication	8
2	Introduction Basics of embedded systems design Embedded OS - Design constraints for mobile applications, both hardware and software related Architecting mobile applications user interfaces for mobile applications touch events and gestures Achieving quality constraints performance, usability, security, availability and modifiability.	
Unit 3	IoT Mobile Apps	8
3	IoT Mobile App Development Trends In 2020 - Role of Mobile Apps in revolutionizing the world of IoT - UX / UI design for IoT Mobile apps - challenges of UX/UI design for IoT applications - practice tips on design for IoT mobile apps IoT App Design Solutions	

Unit 4	Physical Devices and Endpoints	8
4	Physical Devices and Endpoints: Arduino Pin diagram, Arduino Architecture, Arduino Programming, Raspberry Pi Pin diagram, Raspberry Pi Architecture, Compatible Peripherals, Add-Ons, Accessories & Interfaces, Operating System for Raspberry Pi, Setting up and Initial Configuration for Raspberry Pi, Sensors and Interfacing: Controlling LED using switch, Types of Sensors, Integrating Sensors: HDT (Humidity and Temperature Sensor), Light Sensor (LDR), Gas Detector, Ultrasonic Sensor, IR Obstacle Sensor, Fire Sensor and others, Wi-Fi Module, Bluetooth Module, GSM, Camera, Intel Galileo Pin Diagram, Architecture and Interfacing and programming.	

List of Practical

1. To study Arduino, Raspberry Pi and Intel Galileo board
2. To configure the boards for interfacing with sensors.
3. Interface, control & program LED using switch.
4. Interfacing & programming Light Sensor
5. Interfacing & programming Humidity and Temperature Sensor
6. Interfacing & programming Gas Detector
7. Interfacing & programming Ultrasonic Sensor
8. Interfacing & programming Obstacle Sensor
9. Interfacing & programming Fire Sensor
10. Interfacing & programming Bluetooth module
11. Interfacing & programming Wi-Fi Module
12. Interfacing & programming GSM
13. Interfacing & programming Camera
14. Web server Creation
15. Computer App Designing with the boards.

Connecting Embedded Boards with Cloud.

Learning Approach:

- To Study the Concepts of IoT & Develop IoT Applications.

References/suggested learning resources:

TextBook and Learning Resource:

- S. K. Vasudevan, A. S. Nagarajan, RMD Sundaram, "Internet of Things", Wiley.
- G. C. Hillar, "Internet of Things with Python", PACKT Publications.
- A. Bahga, V. Madiseti, "Internet of Things: A Hands-on Approach", Orient Blackswan Private Limited - New Delhi
- J. C. Shovic, "Raspberry Pi IoT Projects: Prototyping Experiments for Makers", Apress.
- M. Schwartz, "Internet of things with the Arduino Yun", Packt Publishing Ltd.
- Richard Grimmett, "Intel Galileo Essentials", Packt Publishing.
- O. Hersent, D. Boswarthick, O. Elloumi, "The Internet of Things: Key Applications and Protocols", Wiley.
- Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos, David Boyle, "From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence", 1st Edition, Academic Press, 2014.
- Peter Waher, "Learning Internet of Things", PACKT publishing, BIRMINGHAM – MUMBAI
- Bernd Scholz-Reiter, Florian Michahelles, "Architecting the Internet of Things", ISBN 978-3-642-19156-5 e-ISBN 978-3-642-19157-2, Springer

Reference Books:

- Daniel Minoli, "Building the Internet of Things with IPv6 and MIPv6: The Evolving

- World of M2M Communications”, ISBN: 978-1-118-47347-4, Willy Publications
- Vijay Madisetti and Arshdeep Bahga, “Internet of Things (A Hands-on Approach)”, 1st Edition, VPT, 2014.
 - C. Dierbach, “Introduction to Computer Science using Python: A Computational Problem-Solving Focus”, John Wiley & Sons.

(b) Open source software and website address:

- http://www.cse.wustl.edu/~jain/cse570-15/ftp/iot_prot/index.html
- Swayam Course on Introduction To Internet Of Things, By Prof. Sudip Misra, IIT Kharagpur
- Swayam Course on Introduction To Industry 4.0 And Industrial Internet Of Things By Prof. Sudip Misra, IIT Kharagpur

Course Name: Introduction to DevOps Technology	
Semester V	
Course Code:	BCA-EC506
Course Title:	Introduction to DevOps Technology
Number of Credits:	Total: (L: 2 T:1 P: 2)
Total Number of teaching hours	40 Hours
Pre-requisites	NIL
Course Category:	Elective

Rationale:

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcome:

- This course covers best practices, tools, and techniques for automating tasks and collaborating with other engineers on projects.

Detailed Contents

Unit	Title	Hours
Unit 1	Overview of DevOps	10
1.1	Why DevOps, What is DevOps, DevOps Market Trends DevOps Engineer Skills, DevOps Delivery Pipeline, DevOps Ecosystem	
1.2	What is version control, What is Git, Why Git for your organization, Install Git, Common commands in Git, Working with Remote Repositories	
Unit 2	Git, Jenkins & Maven Integration & Continuous Integration using Jenkins	10

2.1	Branching and Merging in Git, Git workflows, Git cheat sheet, What is CI, Why CI is Required, Introduction to Jenkins (With Architecture), Introduction to Maven	
2.2	Jenkins Management, Adding a slave node to Jenkins, Building Delivery Pipeline, Pipeline as a Code, Implementation of Jenkins, Build the pipeline of jobs using Jenkins, Create a pipeline script to deploy an application over the tomcat server	
Unit 3	Continuous Deployment: Containerization with Docker	10
3.1	Introducing Docker, Understanding images and containers, Running Hello World in Docker, Introduction to Container, Container Life Cycle, Sharing and Copying Base Image, Dockerfile Working with containers, Publishing Image on Docker Hub	
3.2	Introduction to Docker Ecosystem, Docker Compose, Docker Swarm, Managing Containers, Running Containers, Introduction to Docker Networking, Network Types, Docker Container Networking	
Unit 4	Introduction to DevOps on Cloud	10
4.1	Why Cloud, Introduction to Cloud Computing, Why DevOps on Cloud, Introduction to AWS Various, AWS services, DevOps using AWS	

List of Practical:

As per the Subject teacher assigned as per theory curriculum.

Learning Approach:

The goal of this syllabus is to give you an overview of the skills needed for success in this role so that when a student is looking for work or interviewing for positions, he will be able to confidently say Yes, I know how to do that.

1. (a) Books

1. *The DevOps Handbook: How to Create World-Class Agility, Reliability & Security in Technology Organizations*, written by Gene Kim, Jez Humble, Patrick Debois, and John Willis, was published by IT Revolution Press in 2016.

2. *Effective DevOps: Building a Culture of Collaboration, Affinity, and Tooling at Scale* by Jennifer Davis and Ryn Daniels was also released in 2016. It was published by O'Reilly Media, Inc.
3. *The DevOps Adoption Playbook: A Guide to Adopting DevOps in a Multi-Speed IT Enterprise* by Sanjeev Sharma was released in 2017. It was published by Wiley.